**ABISHEK DEVARAJ(192512126)**

**Experiment: 10**

# 5G Network Slicing Environment Using MATLAB

## Aim

To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

## Objectives

**Objective – 1: Analyze the Number of Network Slices**

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

**Objective – 2: Compare Slice Utilization (%)**

Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

**Objective – 3: Analyze End-to-End Latency (ms)**

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

## Software Required

- **MATLAB**

## Theory

Network slicing is a key feature of 5G that enables a single physical network infrastructure to be partitioned into multiple logical networks called slices. Each slice is optimized for a specific service category, such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), or massive Machine-Type Communications (mMTC). By dynamically allocating network resources, network slicing improves flexibility, resource utilization, and Quality of Service (QoS) while supporting diverse application requirements.

## Objective – 1

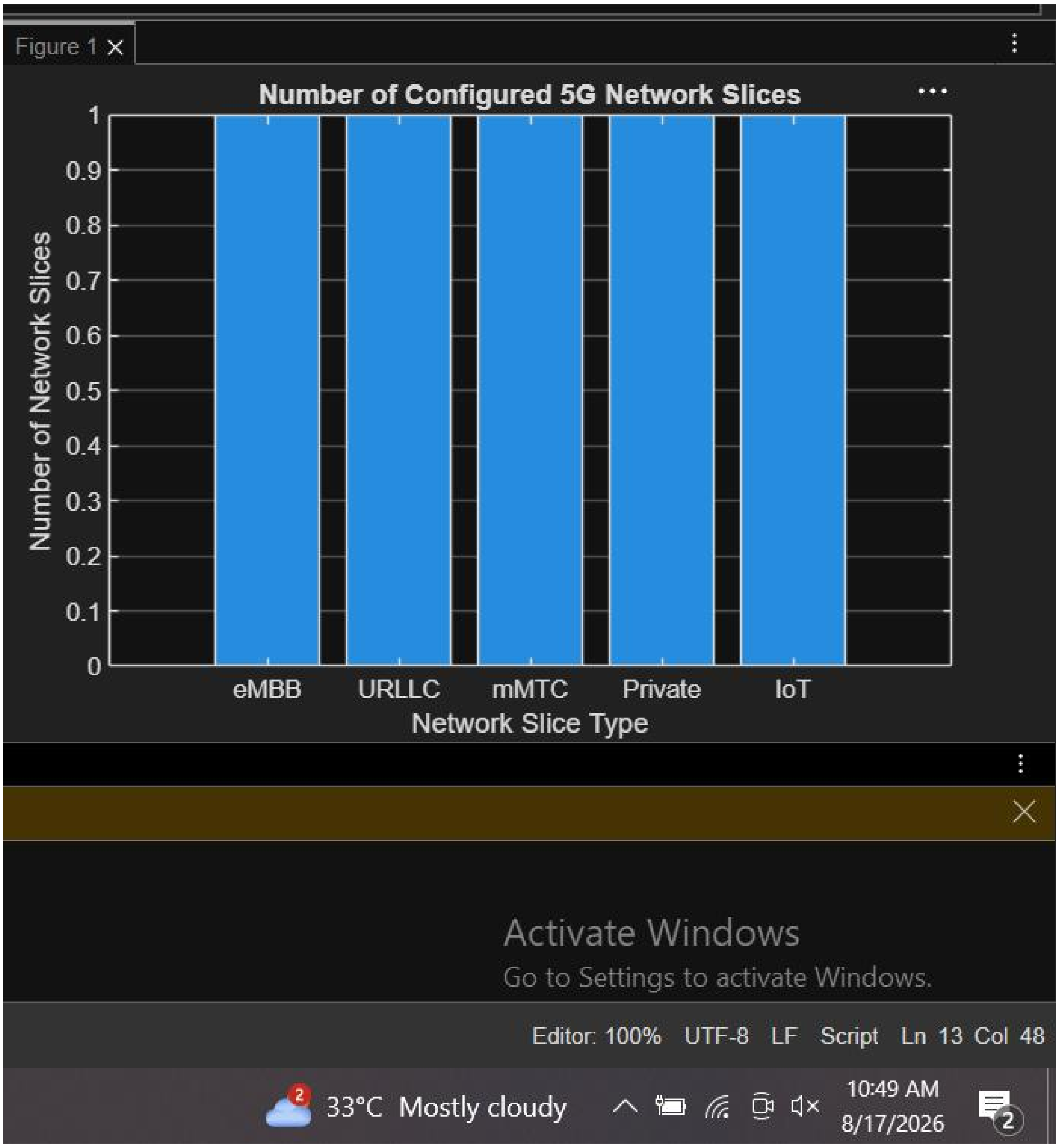
**Analyze the Number of Network Slices**

Generate and visualize multiple network slices using MATLAB. Observe how a 5G network can be logically partitioned into independent slices for different communication services.

**MATLAB Code:**

```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};count = [1 1 1 1 1]; % One instance of each network slice
figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```

**Expected**




• **Bar graph showing the number of configured network slices.**

**Objective – 2**

**Compare Slice Utilization (%)**

Analyze the utilization percentage of different network slices using MATLAB. Compare the

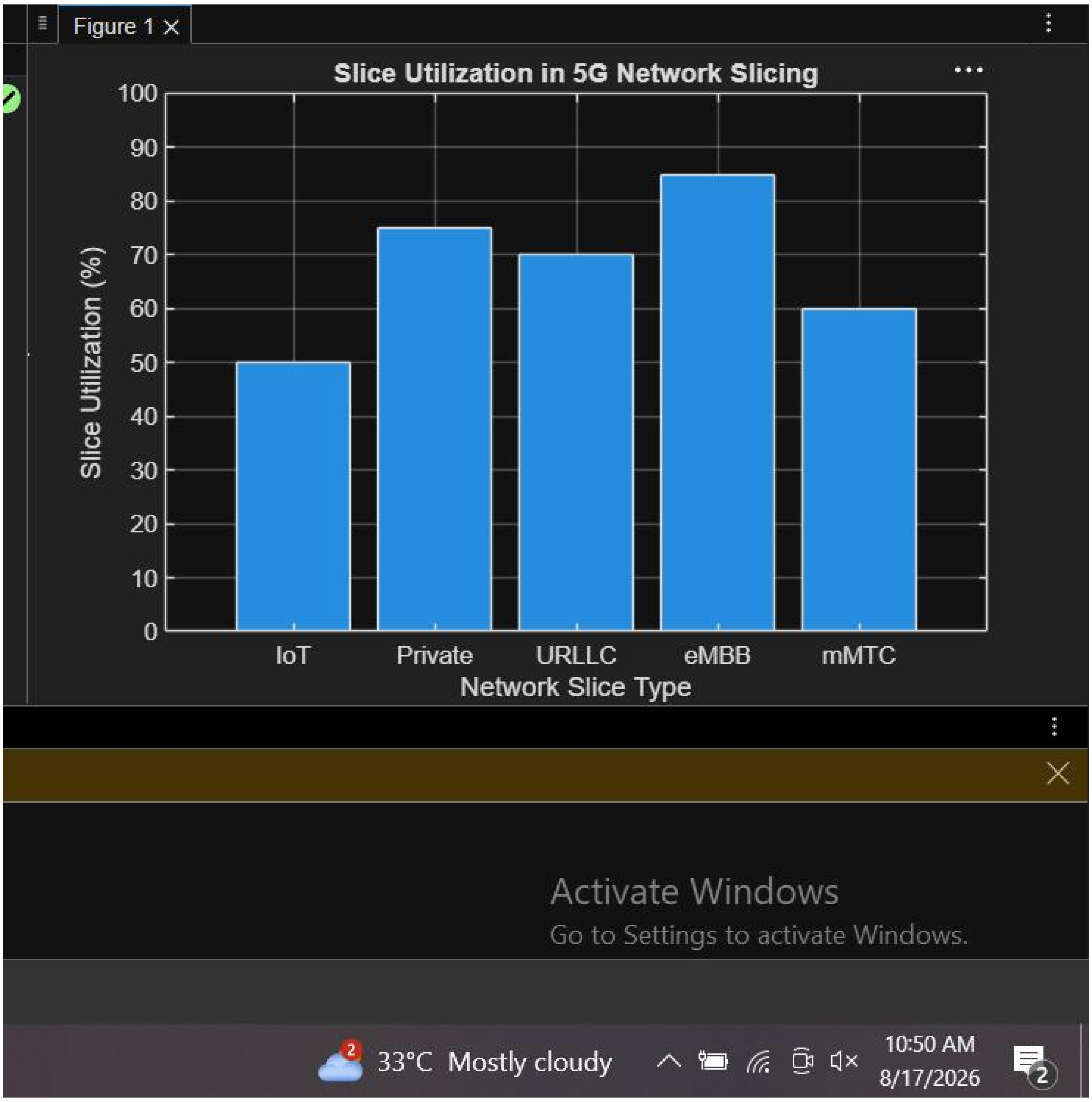
resource usage of each slice to evaluate network efficiency.

**MATLAB Code**

```
clc;
```

```
clear;
close all;
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];figure;
bar(slices, utilization);
grid on;
xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title('Slice Utilization in 5G Network Slicing');
ylim([0 100]);
```

## Expected Output




• **Bar graph showing slice utilization (%).**

**Objective – 3**

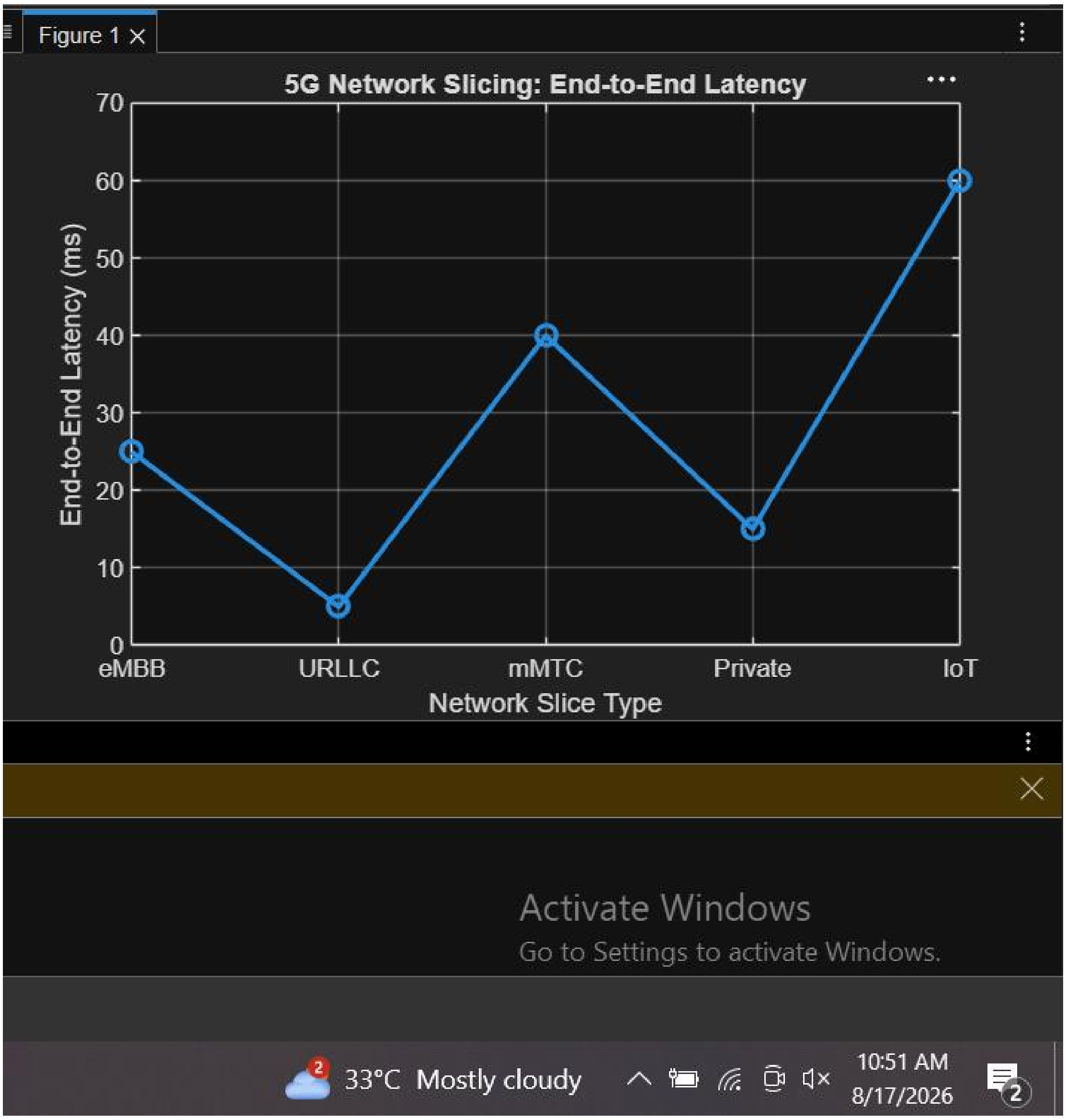
**Analyze End-to-End Latency (ms)**

Compare the end-to-end latency of different network slices using MATLAB. Observe how latency varies depending on the service requirements of each slice.

**MATLAB Code:**

```
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);
```

**Expected Output**




• **Line graph showing end-to-end latency for different network slices.**

**Result**

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting